FORDHAM
THE JESUIT UNIVERSITY OF NEW YORK

Gabelli School of Business

Lecture IV – Accuracy

Machine Learning for Business

Michael Deamer

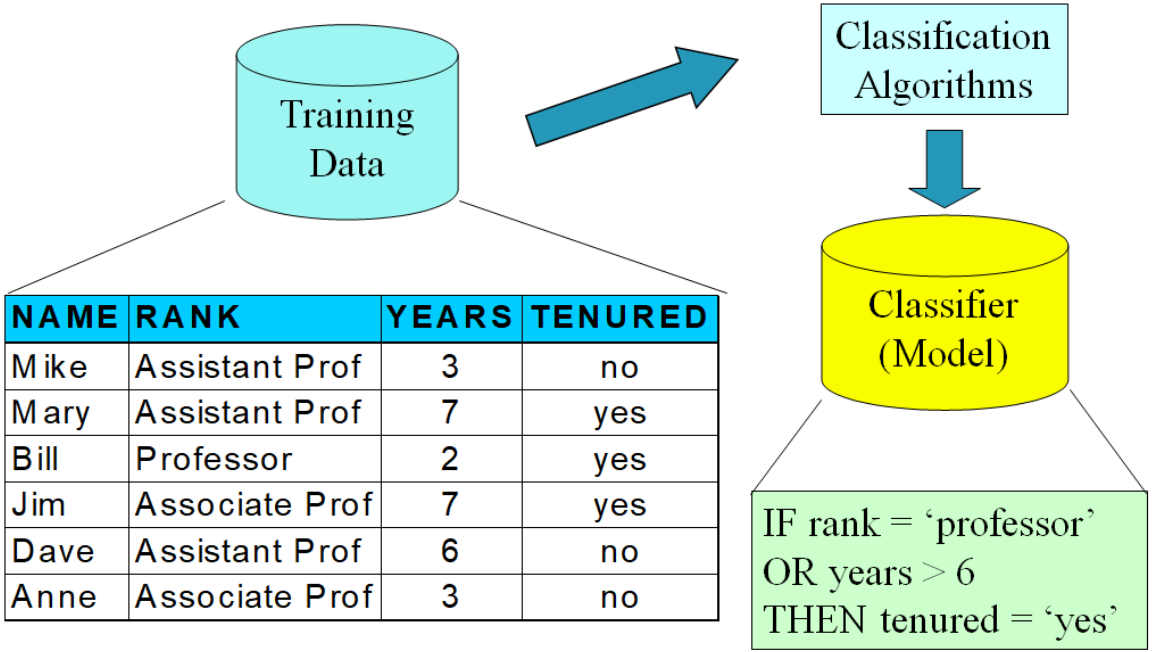
Classification

Prediction

Surface	Mouth	Nose	Voice	Locomotion	Class
Skin	Oval	Snout	Squeek	Walk	Pig
Feathers	Bill	Holes	Quack	Waddle	Duck
Skin	Oval	Aquiline	Twitter	Automobile	Human
Feathers	Bill	Holes	Quack	Swims	?

Classification Process

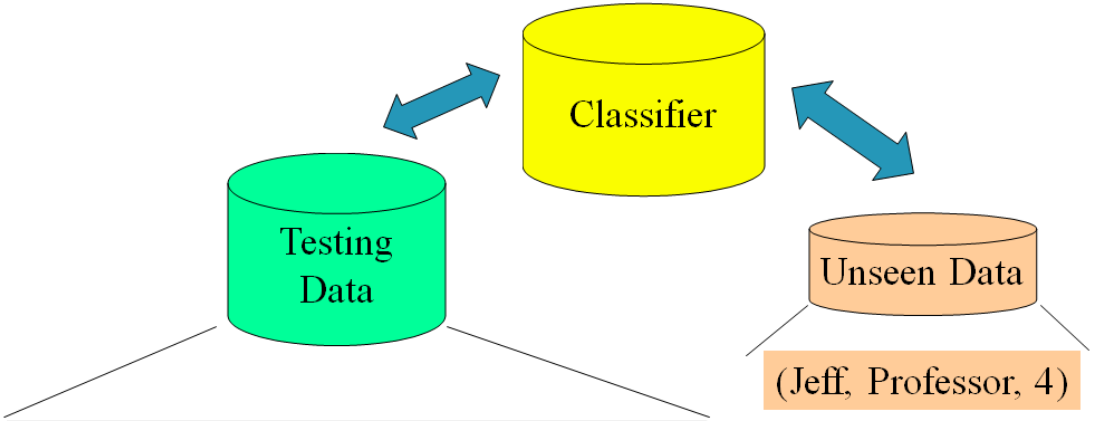
Step I – Model Construction



NAME	RANK	YEARS	TENURED
Mike	Assistant Prof	3	no
Mary	Assistant Prof	7	yes
Bill	Professor	2	yes
Jim	Associate Prof	7	yes
Dave	Assistant Prof	6	no
Anne	Associate Prof	3	no

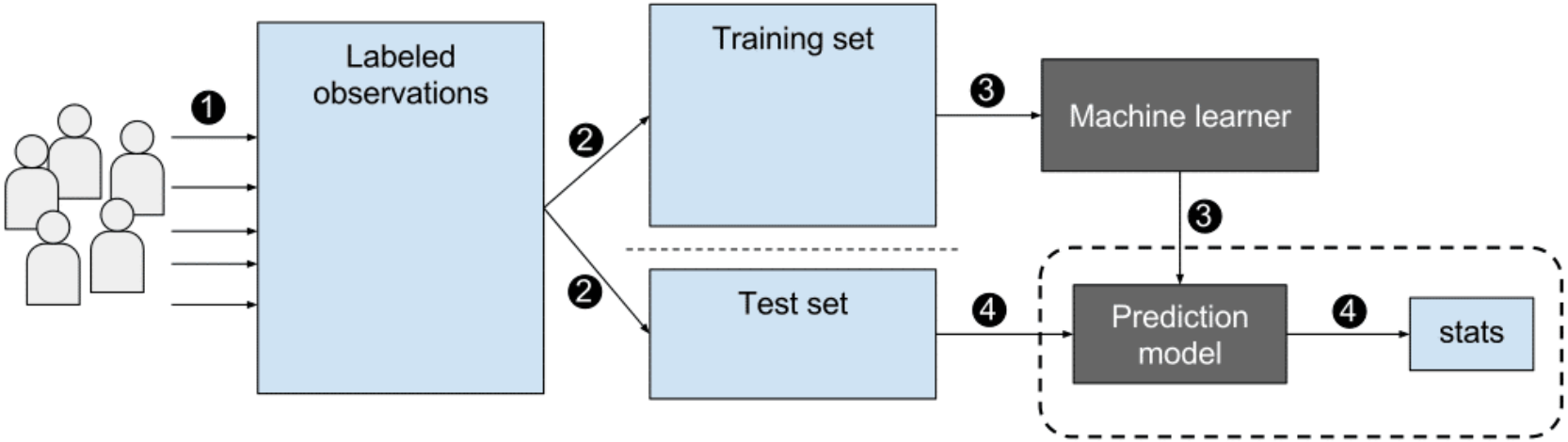
Classification Process

Step II – Test Accuracy of Model



NAME	RANK	YEARS	TENURED
Tom	Assistant Prof	2	no
Merlisa	Associate Prof	7	no
George	Professor	5	yes
Joseph	Assistant Prof	7	yes

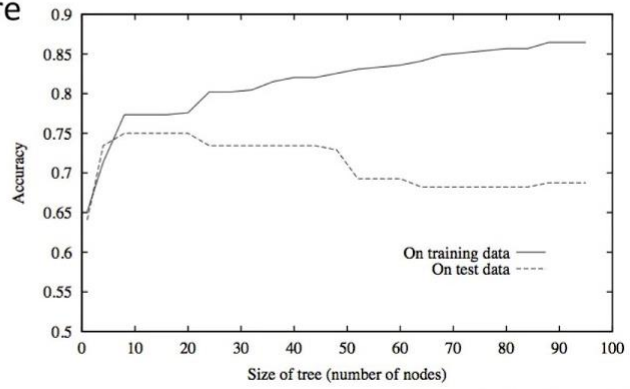
Training – Testing



Identifying Overfitting

Overfitting in Decision Trees

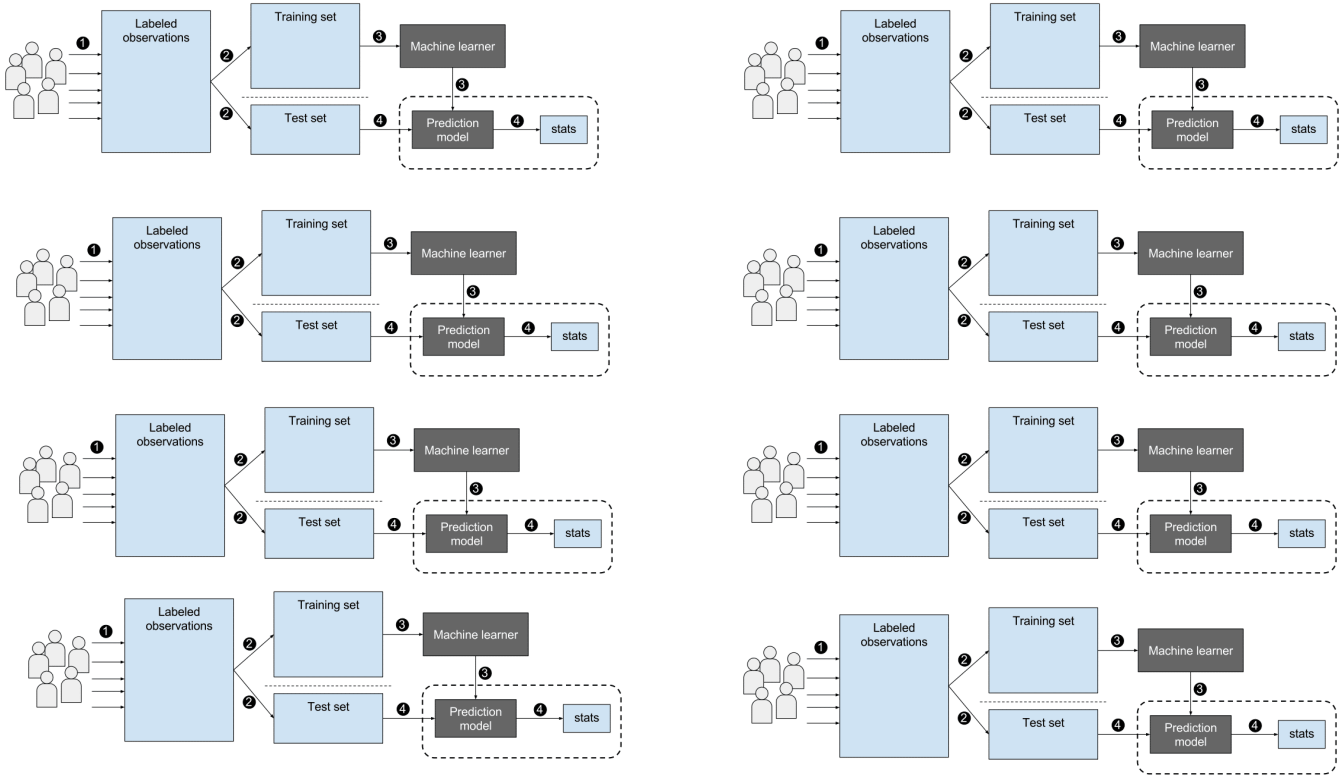
- Can always classify training examples perfectly
  - keep splitting until each node contains 1 example
  - singleton = pure
- Doesn't work on new data



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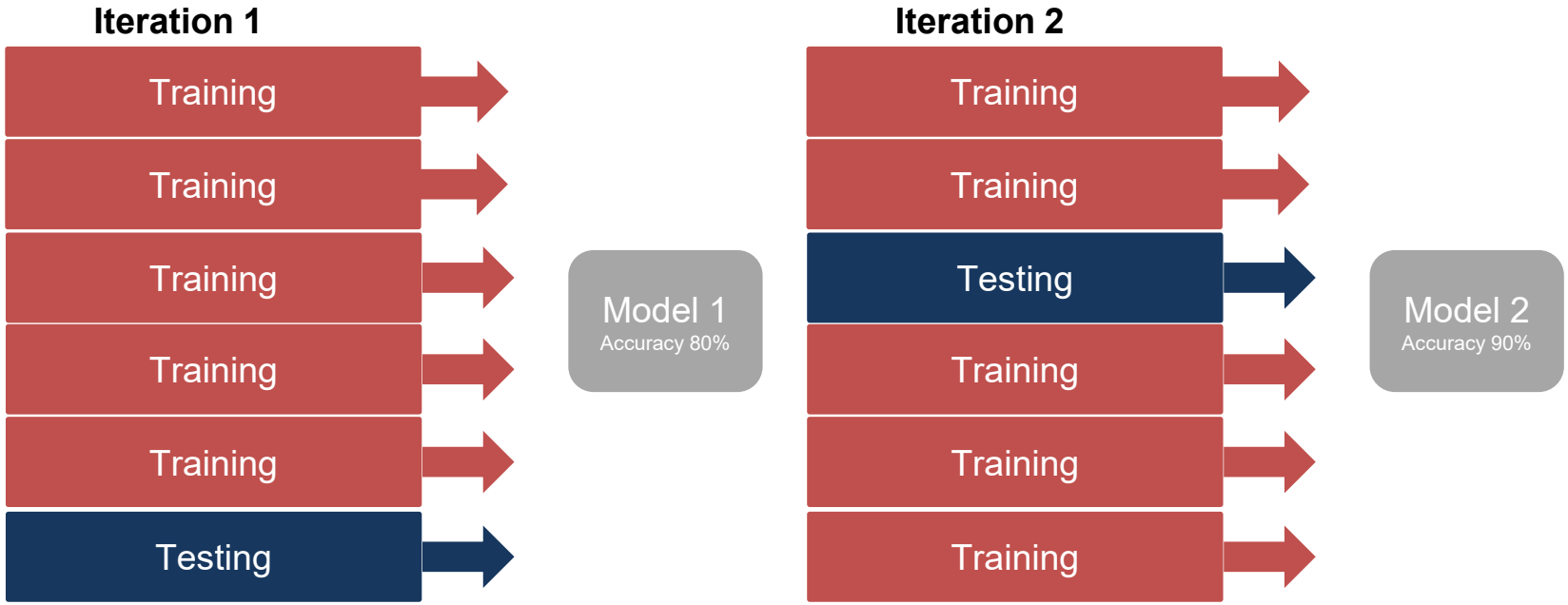
Figure credit: Tom Mitchell, 1997

Ensembles


Labeled observations
Training set
Machine learner
Test set
Prediction model
stats

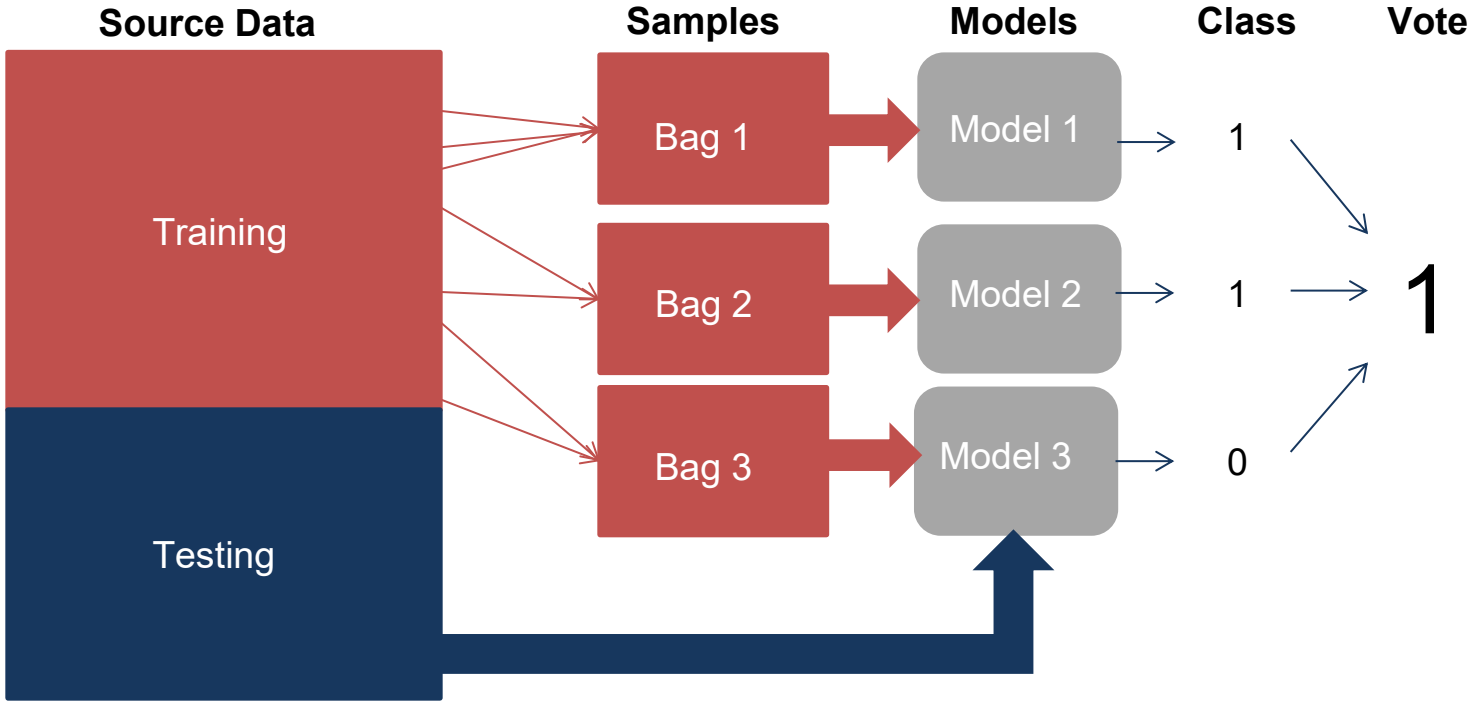
Ensembles

Cross Validation



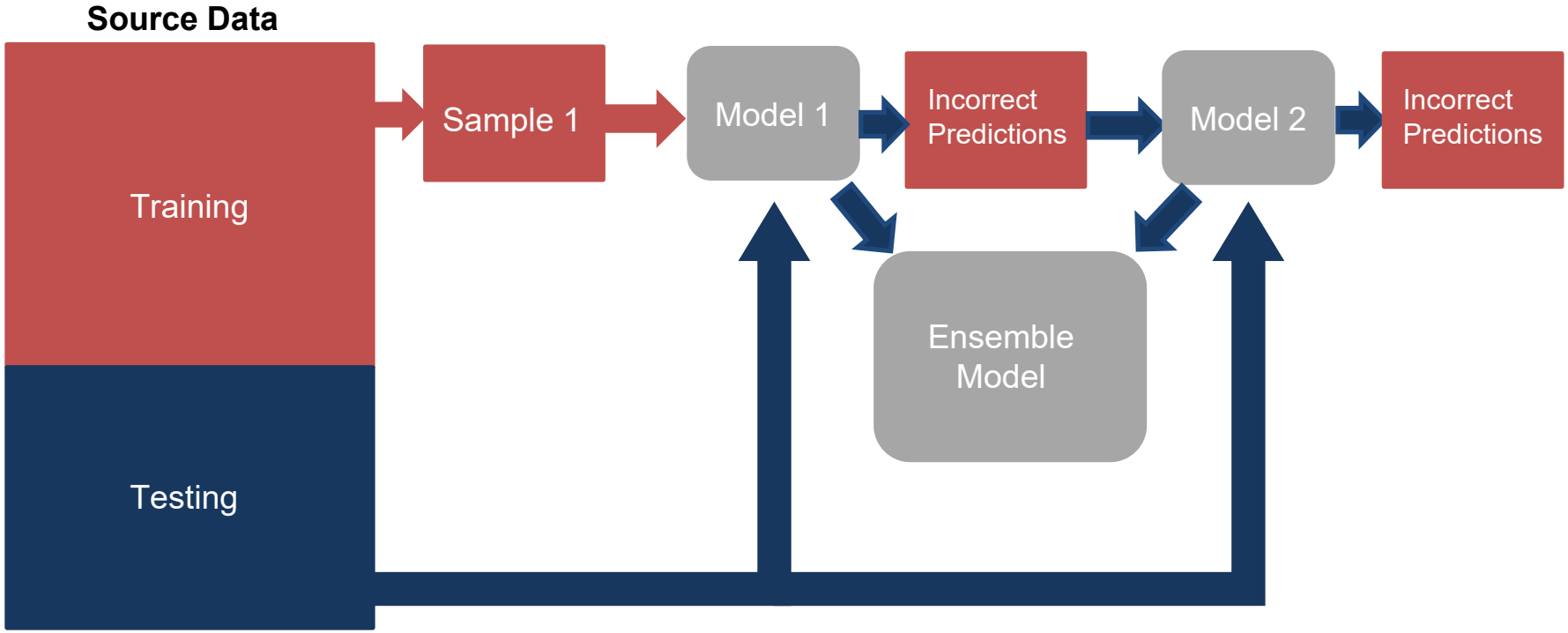
Ensembles

Bagging: Bootstrap Aggregation



Ensembles

Boosting



To Do

- ❑ Lab III
- ❑ Homework III
- ❑ Recommended reading: section 6.5

Bibliography

Tan, Steinbach, Karpatne, and Kumar. Introduction to Data Mining, second edition. Pearson, Hudson, NY, 2019.
Han, Kamber, Pei. Data Mining Concepts and Techniques, second edition. University of Illinois at Urbana-Champaign. Elsevier, San Francisco, CA, 2006.